

MATH 1342 – Elementary Statistics – Week 11: Normal Distributions and Inference for a Proportion

Nov 3 – 5 | 5.1 Hypothesis Testing Using Normal Distributions • 6.1 Inference for a Proportion (Distribution, CI, Hypothesis Test) | Fall 2026 · Section 036 · TTh
6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Explain when the Central Limit Theorem justifies replacing a simulated randomization distribution with a normal distribution $N(\text{null value}, SE)$. (CO4, CO8) 2. Compute a standardized z statistic and find the p-value as a right-tail, left-tail, or two-tail area of the standard normal distribution. (CO4, CO10) 3. Check the large-sample condition (at least 10 in each category) for inference about one proportion. (CO8) 4. Construct and interpret a confidence interval for a population proportion with z^* for 90%, 95%, or 99% confidence. (CO4, CO9) 5. Carry out a one-proportion z test and conclude in context, and explain why the CI standard error uses $\hat{p}$ while the test uses p_0. (CO4, CO10)	e-Text: 5.1 Hypothesis Tests Using Normal Distributions	MO1, MO2	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value	MO1, MO2	Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 8, 11:59 PM)	MO1, MO2
		Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions	MO1, MO2				
		Slides: 5.1 Hypothesis Tests Using Normal Distributions	MO1, MO2	Participation: Inference for a Proportion – 4-step CI and 4-step test for a proportion (voting example); compare the two SEs	MO3, MO4, MO5	Online Homework: 6.1 Inference for a Proportion (due Sun Nov 8, 11:59 PM)	MO3, MO4, MO5
		e-Text: 6.1 Inference for a Proportion (6.1-D Distribution, 6.1-CI Confidence Interval, 6.1-HT Hypothesis Test)	MO3, MO4, MO5			Test 2 Retake – Mon Nov 2 – Sun Nov 8 (optional; higher score counts)	None
		Interactive Lecture: 6.1 Inference for a Proportion	MO3, MO4, MO5				
		Slides: 6.1 Inference for a Proportion	MO3, MO4, MO5				
		StatKey – Theoretical Distributions (Normal)	MO2, MO4				